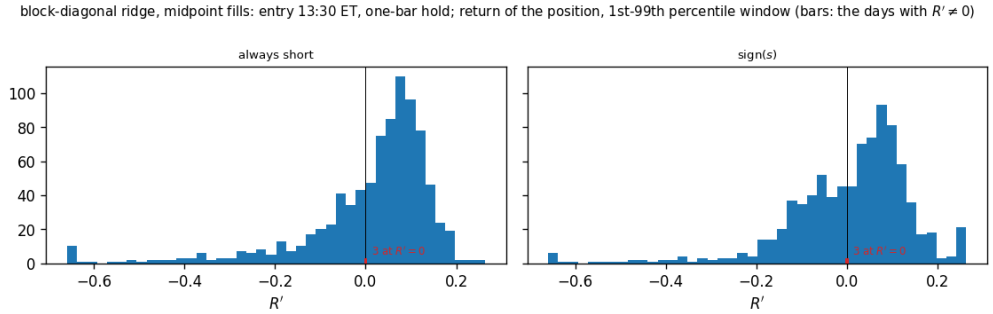
**0DTE nearest-OTM straddle: per-rule return summary, entry 13:30 ET, one-bar hold, models compared on the same 866 days**

| | $n$ | mean | std | min | max | skew | ex. kurt | $t$ | $\text{Sharpe}_{\text{clk}}$ | $n_{\text{buy}}$ | %buy | $\text{Sharpe}^{\times}_{\text{clk}}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Short volatility (always short)* | | | | | | | | | | | | |
| all models (no forecast) | 866 | 0.008 | 0.161 | -1.07 | 0.25 | -2.85 | 11.71 | 1.46 | 0.79 | 0 | 0.0 | -3.47 |
| sign($s$) (±1 *on the sign of* $s$) | | | | | | | | | | | | |
| baseline (HAR + calendar OLS) | 866 | 0.009 | 0.161 | -1.07 | 0.96 | -1.25 | 11.46 | 1.71 | 0.92 | 189 | 21.8 | -3.40 |
| block-diagonal ridge | 866 | 0.008 | 0.161 | -1.07 | 0.96 | -1.40 | 11.44 | 1.51 | 0.82 | 205 | 23.7 | -3.52 |
| block-diagonal ridge, without the FOMC columns | 866 | 0.009 | 0.161 | -1.01 | 1.07 | -0.75 | 11.34 | 1.71 | 0.92 | 215 | 24.8 | -3.42 |
| LightGBM† | 866 | 0.005 | 0.161 | -1.07 | 0.85 | -2.01 | 11.34 | 0.85 | 0.46 | 193 | 22.3 | -3.82 |
| XGBoost† | 866 | 0.006 | 0.161 | -1.07 | 0.85 | -1.90 | 11.38 | 1.02 | 0.55 | 197 | 22.7 | -3.73 |
| lasso (causally tuned)† | 866 | 0.005 | 0.161 | -1.07 | 0.96 | -1.38 | 11.28 | 0.89 | 0.48 | 200 | 23.1 | -3.83 |
| lasso ($\alpha = 10^{-4}$) | 866 | 0.008 | 0.161 | -1.07 | 0.96 | -1.40 | 11.42 | 1.44 | 0.78 | 205 | 23.7 | -3.56 |
| elastic net (causally tuned)† | 866 | 0.005 | 0.161 | -1.07 | 0.96 | -1.45 | 11.29 | 0.91 | 0.49 | 194 | 22.4 | -3.81 |




Distribution of the return of the position taken at 13:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged. **Straddle** here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every $t$ here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean/std}) \times \sqrt{252}$ of that clock's series: one 30-minute hold per expiration day. It answers "if I only entered at this clock, what is my annual Sharpe?" It is *not* the book's $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ of $R_d^{day} = \sum_t R'_{d,t}$, the sum of the day's twelve holds. That reserved name is the book's daily P&L after pooling the session. At 15:30 the two coincide (one trade that day); they do not on 10:00–15:00 or on the pooled page. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. $n_{\text{buy}}$ is the number of expiration days with position $> 0$ (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook's column. The other column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}^2_{\text{hr}} h_t w_t$: the forecast against the implied variance of the same window. $w_t$ is the expanding mean of each prior day's remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining $h_t$ hours — so every bar carries a signal and no bar sits flat.